

Yacin Ben Kacem

Software Engineering Student - Agentic AI & Backend Systems

yassine.benkacem@eniso.u-sousse.tn | +216 29 504 806 | [linkedin.com/in/yacin-ben-kacem](https://www.linkedin.com/in/yacin-ben-kacem) | github.com/YACINBK | yacinbk.github.io

SUMMARY

Second-year Software Engineering student at ENISO focused on agentic AI systems, secure backend orchestration, and automation pipelines designed for real integration constraints. Builds AI workflows that connect LLMs, RAG, OCR/image-processing services, and tool-calling agents with production-minded backends, identity, caching, containerization, and fallback-aware delivery.

EXPERIENCE

White Cape Technologies - Software Engineering Intern

Jun 2025 - Jul 2025

Sousse, Tunisia

- Architected and delivered **UX Insight Platform**, an AI-driven distributed platform for automated web application UX analysis combining behavioral telemetry, screenshots, OCR, and visual heuristics.
- Engineered Spring Boot backend services with Keycloak-based OAuth2/OIDC security, JWT validation, and access-control checks to support secure delivery and protected platform workflows.
- Orchestrated fallback-aware AI workflows through isolated Python/FastAPI microservices for LLM inference, OCR, and image-processing tasks, reducing coupling between AI workloads and core backend services.
- Built resilient data pipelines for high-volume JSON interaction logs and OCR payloads, backed by PostgreSQL, Redis caching, and Docker Compose.

PROJECTS

QuickFlow - AI Productivity Assistant

2025

- Built a full-stack AI productivity assistant for meeting summaries, structured minutes, and professional email generation.
- Developed Spring Boot backend services and Angular frontend workflows; integrated secure identity management with **Keycloak** and social login support.
- Designed reusable communication automation flows for business writing and meeting documentation, keeping authentication and workflow reliability as first-class concerns.

AI Image Vectorization Pipeline

2025

- Designed a Python-based AI image generation and vectorization pipeline for apparel design workflows under 8GB VRAM constraints.
- Automated preprocessing, segmentation/background removal, color simplification, and SVG tracing to produce repeatable vector-ready outputs.
- Treated hardware limits and repeatability as design constraints for reliable generation-to-vector workflows.

Machine Learning / AI Data Projects

2024 - 2025

- Play Store Predictive Analysis:** trained deep-learning models on a 3M-row dataset to predict app installs and extract application performance insights.
- Soil Nutrient Prediction:** built a reproducible Python pipeline for satellite/environmental data cleaning, feature engineering, and multi-output LightGBM modeling.

SKILLS

Programming: Python, Java, C/C++, C#, TypeScript/JavaScript, SQL

Agentic AI & LLMs: LangChain, LangGraph, MCP servers, tool-calling agents, RAG pipelines, prompt engineering, LLM workflow orchestration

Machine Learning & Data: NumPy, pandas, Pydantic, scikit-learn, PyTorch, Keras, data preprocessing, feature engineering

Image Processing & Vectorization: Segment Anything (SAM), background removal, VTracer/SVG tracing, Oklab color quantization, OpenCV/Pillow preprocessing, batch vectorization workflows

Backend & Security: Spring Boot, FastAPI, Flask, Keycloak, OAuth2/OIDC, JWT, PostgreSQL, Redis

Frontend & AI Apps: Angular, React, Streamlit, Gradio

Infrastructure & Delivery: Docker, Docker Compose, Linux CLI, Git/GitHub/GitLab, Postman, caching, fallback-aware workflows

AI Security: NVIDIA Morpheus fundamentals, AI cybersecurity pipelines

CERTIFICATIONS

NVIDIA: Prompt Engineering for LLMs · RAG Agents with LLMs · AI Cybersecurity Pipelines · Transformer NLP · Diffusion Models · Real-Time Video AI · Predictive Maintenance · Fundamentals of Deep Learning

Google Cloud: Introduction to Generative AI

DeepLearning.AI: Neural Networks and Deep Learning · Supervised Machine Learning

365 Data Science: Data Cleaning and Preprocessing with pandas

EDUCATION

École Nationale d'Ingénieurs de Sousse (ENISO)

Sep 2024 - Present

Engineering Degree in Software Engineering

Preparatory Institute for Engineering Studies of Monastir

Sep 2022 - Jun 2024

Mathematics & Physics Preparatory Cycle, National Rank: 509 / 1800

LANGUAGES

Arabic: Native | English: C1 | French: B2 | German: A1